LS3202 Biostatistics
Mid-Semester Examination 2026 – Solved

(16th February 2026)

Officially solved paper, typeset for student reference

Note on this paper

Total Marks: 30. Instructions: All steps must be shown clearly; formulae used must be written down clearly, with variables named. This paper was provided **already solved with an official marking scheme** (marks shown in **red** throughout, exactly as annotated on the original). This document is a clean LaTeX typesetting of that solved paper, with all numerical values cross-checked. The original scanned PDF is kept alongside this file in `question_papers/` for reference.

Contents

1 Question 1 – Binomial process, Poisson approximation (Marks: 2+2+2 = 6)

Q1

In a city, chances of rain on any day in November is 10%.

i) Why is this a Binomial process? Thereby, what are the chances that November 2026 will experience 5 days of rain?

ii) Treat the same as a Poisson process.

iii) Is the Poisson approximation appropriate in this case? Comment.

Solution.

i) Binomial process

This is a **Binomial process with 2 kinds of events: Rain, No Rain.** **[$\frac{1}{2}$ mark]**

$$\text{Prob. of rain, } s = 10\% = 0.1$$

$$\text{Prob. of no rain} = 0.9$$

$$\text{Total no. of days} = 30, \qquad \text{No. of days of rain, } n = 5$$

Using the **Binomial Distribution**:

$$P(N; n, s) = {}^{N}C_n \, (s)^n (1-s)^{N-n}$$

$$= {}^{30}C_5 \, (0.1)^5 (0.9)^{25} \textbf{[1 mark]}$$

$$= (142506) \times (10^{-5}) \times (7.17897 \times 10^{-2})$$

$$= 10230.477 \times 10^{-5}$$

$$\boxed{= 0.1023} \textbf{[}\frac{1}{2}\textbf{ mark]}$$

ii) Poisson process

Now, $N \cdot s = 30 \times 0.1 = 3 \ll N$. So, we may try the **Poisson distribution**.

For a mean no. of events λ, the probability of 'n' successes is given as:

$$P(n, \lambda) = \frac{\lambda^n e^{-\lambda}}{n!}$$

Hence, $\lambda = 30 \times 0.1 = 3$, and $n = 5$. **[$\frac{1}{2}$ mark]**

$$P(5, 3) = \frac{3^5 e^{-3}}{5!} = (243) \times (0.04978706)$$

$$\boxed{= 0.10081 \approx 0.1008} \textbf{[1 mark, } \frac{1}{2} \textbf{ mark]}$$

iii) Is the Poisson approximation appropriate?

Yes.

$$N \geq 100 \quad \text{(rule of thumb: large } N) \qquad p \leq 0.05 \quad \text{(rule of thumb: small } p)$$

The agreement between the Binomial (0.1023) and the Poisson (0.1008) process is good. **The Poisson approximation holds.** **[2 marks]**

Note: Marks were given for logical explanations for both the answers (parts i and ii), not just the final numbers.

2 Question 2 – One-sample t-test: mangrove growth rate (Marks: 2+1+4+1 = 8)

Q2

A scientist studies the growth rate of mangroves. He traced the increase in length, in cm, of five saplings over one year in a location in India. His recorded data was $\{52.5, 48.8, 44.6, 55.5, 45.6\}$. Globally, the mean height increase is known to be 53.0 cm in one year.

i) Record the sample statistics.

ii) State hypotheses to evaluate if this cohort has lower growth rate than the global reference value.

iii) Evaluate your hypothesis with a suitable test. State why you chose this particular test.

iv) Schematically depict the relevant distribution, marking the "acceptability region".

Solution.

i) Sample statistics

$$n = 5, \qquad \nu = 4 \text{ (degrees of freedom)}$$
$$\text{Mean, } \bar{x} = 49.4 \text{ cm} \qquad \text{Variance, } s^2 = 21.165 \textbf{[1 mark]}$$
$$\text{Std. dev., } s = 4.60054$$
$$\text{Std. error, } SE = \frac{s}{\sqrt{n}} = 2.057424 \textbf{[1 mark]}$$

ii) Hypotheses

Reference level is $\mu_0 = 53.0$ cm. Constructing the hypothesis:

$$H_0 : \mu = \mu_0$$
$$H_A : \mu < \mu_0 \textbf{[1 mark]}$$

iii) Test and evaluation

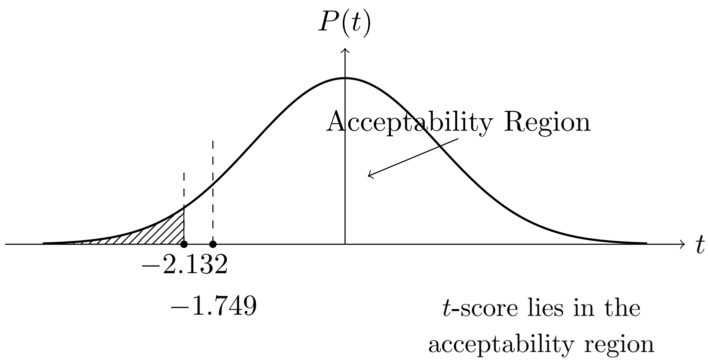
One-tailed t-test, due to small sample size. $CI = 95\%$, $\alpha = 0.05$.

$$t_{score} = \frac{\bar{x} - \mu_0}{SE} \textbf{[1 mark]}$$
$$= \frac{49.4 - 53.0}{2.057424} \textbf{[1 mark]}$$
$$\boxed{t_{score} = -1.749760 \approx -1.750} \textbf{[1 mark]}$$

We will compare $|t_{score}|$ to $t^{(1)}_{0.05,4} = 2.132$. **[$\frac{1}{2}$ mark]**

$\Rightarrow$ Since $|t_{score}| = 1.750 < t_{crit} = 2.132$, **we retain H_0.** **[$\frac{1}{2}$ mark]**

iv) Schematic of the distribution



[1 mark]

3 Question 3 – One-sample Z-test: weight change in cancer patients (Marks: 2+6 = 8)

Q3

A large sample of 100 cancer patients is randomly selected to find the weight change in 2 weeks from the intake of a food supplement. The mean weight change was +1.2 kg, and std. dev. was 0.8 kg. At the population level, the mean weight change of cancer patients is −0.5 kg (in the same time).

i) State hypotheses to evaluate if weight change of this sample is *different* from the reference.

ii) Evaluate your hypothesis at a confidence interval of 95%.

Solution.

i) Hypothesis

$$H_0 : \mu = \mu_0 \ (-0.5)$$

$$H_A : \mu \neq \mu_0 \ (-0.5)$$**[2 marks]**

Note (from original marking scheme)

Mentioning any value other than −0.5 leads to a $-\frac{1}{2}$ mark deduction. Not mentioning any value is okay.

ii) Sample statistics and test

$n = 100\ (\geq 30)$, mean $\bar{x} = 1.2$ kg, std. dev. $s = 0.8$ kg, var $s^2 = 0.64$ kg^2

$$SE = \frac{s}{\sqrt{n}} = \frac{0.8}{\sqrt{100}} = \frac{0.8}{10} = 0.08$$**[2 marks]**

We wish to evaluate whether this sample is different from the population, reference value $\mu_0 = -0.5$ kg. Since the sample is large ($n \geq 30$), we will use the **Z-test**.

$$Z_{score} = \frac{\bar{x} - \mu_0}{SE} = \frac{1.2 - (-0.5)}{0.08}$$

$$\boxed{Z_{score} = 21.25}$$**[2 marks]**

The CI at 95%, i.e., $\alpha = 0.05$ for a two-tailed Z-test:

$$Z_{\alpha/2} = 1.96$$**[1 mark]**

Now, $|21.25| > 1.96$. **We reject** H_0 (or accept H_A). **[1 mark]**

4 Question 4 – Two-sample t-test (pooled variance): legume protein content (Marks: 2+5+1 = 8)

Q4

A nutritionist is interested in comparing the protein content (arb. units) in two legume species. She tested two random samples, whose statistics are reported below:

	Legume-1	Legume-2
Sample size	8	8
Mean	18.2	14.4
Variance	4.0	3.2

She supposes that at the population level, there is no difference in their mean protein content.

i) State hypotheses for the nutritionist. State the most appropriate test in this case.

ii) Evaluate the hypothesis based on the data provided at $\alpha = 0.05$.

iii) Schematically depict the relevant distribution, marking the "acceptability region".

Solution.

Given:

$$n_1 = n_2 = 8, \qquad \bar{x}_1 = 18.2,\ s_1^2 = 4.0, \qquad \bar{x}_2 = 14.4,\ s_2^2 = 3.2$$

i) Hypothesis and test choice

$$H_0 : \mu_1 = \mu_2$$
$$H_1 : \mu_1 \neq \mu_2 \text{[1 mark]}$$

Most Appropriate Test: Two-Sample t-test with pooled variance **[1 mark]** (appropriate since both sample sizes are small and equal, and the two sample variances are reasonably similar, so a pooled-variance approach is justified.)

ii) Evaluation

Pooled variance:

$$Var = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$$
$$= \frac{7 \times 4.0 + 7 \times 3.2}{14} = 3.6\text{[1 mark]}$$

Standard Error:

$$SE = s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$
$$= 1.897\sqrt{0.28} = 0.9487\text{[1 mark]}$$

t-score:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{SE} = \frac{18.2 - 14.4}{0.9487}$$
$$\boxed{t = 4.01}\text{[1 mark]}$$

Degrees of freedom:

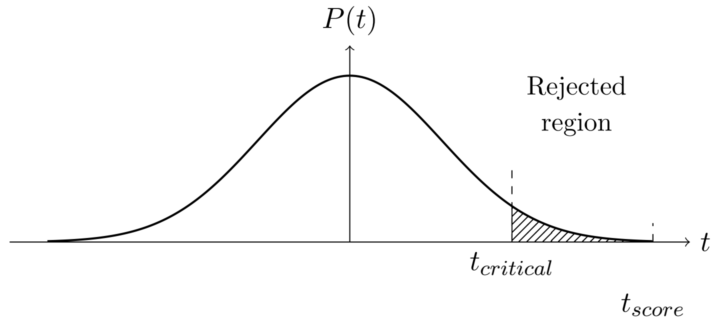
$$\nu = n_1 + n_2 - 2 = 14 \textbf{[1 mark]}$$

$$\alpha = 0.05$$

$$t_{critical} = t_{\alpha/2,\,\nu} = t_{0.025,\,14} = 2.145$$

Since $t_{score} > t_{critical}$, **Reject H_0.** **[1 mark]**

iii) Schematic of the distribution



[1 mark]

5 Formula Sheet

Formulae provided with the paper

$$Z = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}}$$

$$Z_{score} = \frac{[\bar{X}_1 - \bar{X}_2] - [\mu_1 - \mu_2]}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$$

$$t_{score} = \frac{[\bar{X}_1 - \bar{X}_2] - [D_0]}{\sqrt{s^2\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}}$$

$$\text{approx. } \nu \approx \frac{\left[\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}\right]^2}{\dfrac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \dfrac{\left(s_2^2/n_2\right)^2}{n_2 - 1}}$$

$$Var = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 + n_2 - 2)}$$

$$t_{score} = \frac{[\bar{X}_1 - \bar{X}_2] - [D_0]}{\sqrt{\left(\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}\right)}}$$